

Research-Grade Product Diagnosis and Redesign Report for a Blockchain E-Procurement and PLS Financing Seedbed

Prepared for Founder, Product, Design, Engineering, QA, and Delivery Review
Repository context: Blockchain-Based-E-Procurement-System
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Abstract—This report evaluates a blockchain-based e-procurement and profit-loss sharing (PLS) financing seedbed as a potential deployable MVP. The diagnosis finds a defensible market problem at the intersection of procurement transparency, SME financing friction, regulated onboarding, and Shariah governance; however, the current product is stronger as a technical foundation than as an actor-complete commercial workflow. The recommended redesign is a compliance-first procurement evidence platform that uses PostgreSQL for operational state and Hyperledger Fabric only for proof anchoring, while completing a single canonical transaction case across administrator, buyer, supplier, compliance reviewer, Shariah reviewer, financier, auditor, regulator, and platform operator actors. The report includes market validation, product-market fit analysis, UI/UX redesign guidance, technical architecture review, phased implementation roadmap, risks, success metrics, and complete behavioral and structural UML documentation rendered as figures for PDF export.

Index Terms—blockchain, e-procurement, SME finance, Islamic finance, PLS, Hyperledger Fabric, auditability, compliance, product management, UML.

I. EXECUTIVE SUMMARY

The product should not be positioned as a complete blockchain marketplace or production Islamic finance platform. A stronger and more credible position is: *a compliance-first digital procurement and PLS financing seedbed for regulated SME transactions, with tamper-evident audit proof and Shariah-governed financing workflow support*. This is a narrower but more marketable proposition because it directly addresses recurring stakeholder pain: fragmented procurement evidence, limited SME financing readiness, weak auditability, compliance handoff friction, and lack of trustworthy proof during regulator or financier review.

Current product maturity is uneven. The technical foundation is comparatively strong: session-based entry, role-aware dashboard baseline, PostgreSQL persistence baseline, Fabric proof anchoring, blockchain proof UI, audit trail, and escrow first slice are either implemented or evidenced in the repository context. The commercial product layer remains incomplete: several actors still lack visible, usable workflows, especially administrator, supplier, compliance reviewer, Shariah reviewer, financier, regulator, and platform operator. Therefore, the main delivery problem is no longer just technology integration. It is product coherence.

Recommendation: Build one actor-complete canonical demo case before expanding features: Amanah Retail procures from Barakah Supplies, Mabru Finance supports financing, Compliance approves eligibility, Shariah approves PLS structure, Buyer creates escrow, Auditor verifies Fabric proof, and Regulator exports evidence.

II. ASSUMPTIONS AND SCOPE

The following assumptions are used because only limited project context is available.

- 1) **Assumption:** The product is intended first as an academic/final-year MVP and supervisor demo, then as a possible pilot seedbed.
- 2) **Assumption:** The intended MVP remains centralized-first or hybrid: PostgreSQL is the operational database; Fabric is used for selective proof anchoring only.
- 3) **Assumption:** The current repository already contains a working or near-working Sprint 6 baseline for landing/login, buyer/auditor dashboards, proof UI, PostgreSQL baseline, Fabric chaincode baseline, proof gateway, and escrow first slice.
- 4) **Assumption:** Not all actor workflows are complete in the product UI. The redesign therefore focuses on actor completeness rather than adding more blockchain features.
- 5) **Assumption:** The intended first commercial niche is not broad public procurement replacement. It is regulated procurement and SME financing evidence workflow for Islamic finance or development finance contexts.

Out of scope for the MVP are production bank settlement, external ERP certification, full Hyperledger Fabric consortium operations, DID/VC federation, tokenized receivable market operations, automated arbitration, and multi-jurisdiction legal policy automation.

III. MARKET PROBLEM VALIDATION

Procurement is a large, high-risk economic activity. The OECD notes that public procurement expenditure across OECD countries rose from 11.8% of GDP in 2007 to 12.9% of GDP in 2021, and that public procurement is high risk because of the financial interests involved, transaction volume, private-public interaction, and complex value chains [7]. This supports the hypothesis that digital procurement controls, auditability, and risk visibility are meaningful market problems.

Financing friction is also material. Reuters reported an estimated global trade finance gap of USD 2.5 trillion, citing the Asian Development Bank, and highlighted excess demand for trade finance in emerging markets [8]. This supports the product’s SME financing angle, but also implies that a software platform alone will not solve financing unless it improves evidence quality, eligibility decisions, and bank-review confidence.

Islamic finance adds a compliance dimension. The Islamic Financial Services Board describes itself as an international standard-setting organisation promoting the soundness and stability of the Islamic financial services industry, including banking, capital markets, and insurance [11]. This means a PLS workflow cannot be marketed as a generic profit calculator; it must embed governance, review evidence, and Shariah approval traceability.

Blockchain is relevant only if it reduces evidence trust friction. NIST characterizes blockchains as tamper-evident and tamper-resistant distributed ledgers where, under normal operation, a transaction cannot be changed once published [9]. Hyperledger Fabric documentation similarly describes append-only ledger behavior and Fabric’s support for private permissioned networks, smart contracts, and membership-controlled participation [10]. These properties match the product’s audit-proof need, but they do not prove the truth of original input data. Therefore, the MVP should use Fabric for proof anchoring, not as a primary database or marketing magic.

A. Validated Problem Statements

The market problem is real if framed as follows:

- Regulated procurement participants need a reliable chain of evidence across onboarding, orders, escrow, audit, financing, and export.
- SMEs need a stronger evidence package to support procurement-linked financing decisions.
- Islamic finance reviewers need transparent PLS parameter capture, approval history, and distribution traceability.
- Auditors and regulators need tamper-evident event proof without exposing raw sensitive data on-chain.

B. Unvalidated or Weak Claims

The following claims should not be made until later evidence exists:

- The system is a production-ready procurement marketplace.
- The system executes bank settlement or real payment rails.
- The system is a production Hyperledger Fabric consortium.
- The system fully automates Shariah compliance.
- Blockchain alone solves trust in procurement data.

IV. PRODUCT-MARKET FIT ANALYSIS

The strongest early adopter profile is a regulated institution or development-finance program that already manages procurement, SME onboarding, financing review, and audit evidence manually. Examples include Islamic SME financing institutions, procurement-heavy cooperatives, development finance bodies, or pilot regulator sandboxes.

TABLE I
ACTOR JOBS-TO-BE-DONE AND CURRENT GAPS

Actor	Job-to-be-Done	Main Gap
Administrator	Govern organizations, users, and roles.	Admin UI and role workflow incomplete.
Buyer	Create order and escrow with evidence.	Order lifecycle before escrow incomplete.
Supplier	Acknowledge order and provide delivery evidence.	Supplier workflow missing.
Compliance Reviewer	Approve, flag, or block KYC/AML cases.	Product UI and eligibility gating incomplete.
Shariah Reviewer	Review PLS terms and approve/reject.	Visible workflow incomplete.
Financier	Review PLS contract and distribution evidence.	PLS contract and distribution UI incomplete.
Auditor	Verify event proof and inspect history.	Core proof path exists, export path incomplete.
Regulator	Request signed evidence bundle.	Export workflow missing.

A. Jobs-to-be-Done

B. Fit Score

The product has a plausible market wedge but weak commercial readiness. A qualitative score is shown below.

- Problem severity: High.
- Buyer clarity: Medium, because multiple actors dilute the initial wedge.
- Technical differentiation: Medium-high, if proof anchoring is shown honestly.
- UX readiness: Low-medium.
- Compliance credibility: Medium, dependent on KYC and Shariah workflow completion.
- Commercial readiness: Low-medium.

V. RECURRING USER PROBLEMS

Across the product context, recurring user problems are:

- 1) **Fragmented evidence.** Procurement, KYC, escrow, Shariah review, and export evidence are not experienced as one case file.
- 2) **Actor confusion.** The product currently emphasizes technical modules more than human workflows.
- 3) **Trust gap.** Users need to know whether a record is complete, verified, pending, mismatched, or unavailable.
- 4) **Compliance handoff friction.** Eligibility and Shariah decisions must affect downstream actions, not only appear as labels.
- 5) **Overclaiming risk.** Blockchain, escrow, and PLS wording can imply production capabilities that are not actually implemented.
- 6) **Demo fragility.** If runbooks, seeded data, and role UAT scripts are incomplete, the product feels like code rather than deployable software.

VI. PROPOSED PRODUCT SOLUTION

The proposed product is a case-centric procurement evidence workspace. Instead of separate dashboards that feel disconnected, the product should revolve around one canonical transaction case.

A. Canonical Case

Amanah Retail Sdn Bhd procures from Barakah Supplies Sdn Bhd; Mabur Finance Partner participates in a restricted PLS financing seedbed; Compliance reviews onboarding; Shariah reviews PLS terms; Buyer creates the order and escrow; Auditor verifies proof; Regulator exports the evidence package.

B. Value Proposition

A compliance-first procurement and SME financing evidence platform that helps regulated actors on-board participants, execute procurement workflows, anchor proof, review Shariah-governed PLS decisions, and export verifiable audit evidence without exposing sensitive data on-chain.

C. MVP Feature Pillars

- 1) Governed membership and role access.
- 2) KYC/AML eligibility gate.
- 3) Buyer-supplier procurement order workflow.
- 4) Escrow first-slice and lifecycle event emission.
- 5) Blockchain proof visibility and verification.
- 6) Shariah review and PLS contract seedbed.
- 7) Regulator evidence export.
- 8) UAT, deployment, and demo readiness.

VII. BEHAVIORAL UML DIAGRAMS

This section presents behavioral UML views. The diagrams are rendered figures, not merely source code, so they can be directly exported in the PDF.

Fig. 1 shows that the MVP must serve more than buyer and auditor. The product becomes commercially credible only when administrator, supplier, compliance reviewer, Shariah reviewer, financier, regulator, platform operator, and developer workflows exist or are explicitly deferred.

Fig. 2 is the recommended demonstration spine. It solves the current fragmentation problem by making all actor workflows contribute to one reviewable business transaction.

Fig. 3 shows a safe anchoring pattern: the backend records business events, computes a payload hash, attempts anchoring through a gateway, stores proof metadata, and later verifies the proof. This avoids coupling core procurement correctness to Fabric availability.

Fig. 4 highlights an important UX rule: proof states and business states are related but not identical. A valid escrow record may have pending or failed proof, and the UI must not present unavailable proof as verified.

VIII. STRUCTURAL UML DIAGRAMS

This section presents structural views of the redesigned product and architecture.

Fig. 5 emphasizes that the blockchain proof is not the domain itself. It is evidence attached to lifecycle events and export bundles.

Fig. 6 translates abstract classes into a realistic demo case. Seeded objects are essential for supervisor confidence and commercial storytelling.

IX. TECHNICAL ARCHITECTURE REVIEW

A. Strengths

- **Correct hybrid boundary.** PostgreSQL is the operational store; Fabric is proof infrastructure. This matches enterprise blockchain best practice and avoids treating Fabric as a database.
- **Gateway architecture.** A blockchain gateway port protects application logic from direct Fabric SDK coupling.
- **Audit-first structure.** Lifecycle events and payload hashes give the product an evidence foundation.
- **Incremental deployability.** Local Docker PostgreSQL, migration scripts, runbooks, and proof tests support practical demonstration.
- **Contract documentation.** API, session, proof, escrow, and eligibility contracts reduce implementation drift.

B. Weaknesses

- **Actor incompleteness.** The visible product is buyer/auditor-heavy, while the intended business model requires many roles.
- **Runtime composition gap.** PostgreSQL adapters may exist but must be easy to activate in normal demo mode.
- **Fabric operational gap.** Chaincode tests are useful, but live Fabric deployment must be explicitly optional or reliably scripted.
- **PLS credibility gap.** PLS must be controlled by Shariah approval, calculation evidence, and clear scope limits.
- **Export gap.** Regulator-facing export is critical to the audit value proposition but still needs visible workflow completion.

C. Architecture Recommendations

- 1) Use a feature flag or environment variable for PostgreSQL runtime composition.
- 2) Keep in-memory repositories for tests but never let demos silently lose state without explanation.
- 3) Keep Fabric unavailable states explicit; do not block base business events on anchor failure.
- 4) Add a signed or hash-manifest export bundle before claiming regulator readiness.
- 5) Treat PLS calculations as deterministic scenario records, not payment execution.

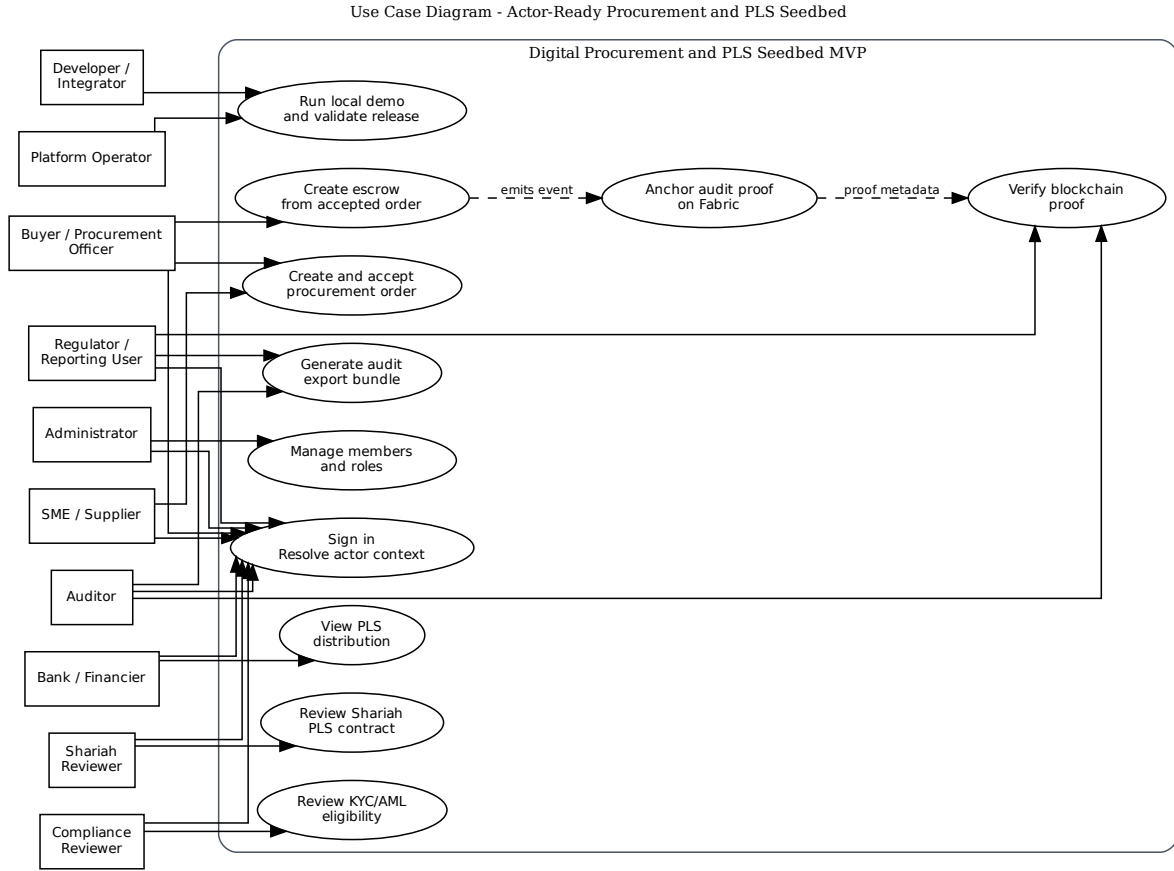


Fig. 1. Use case diagram for the actor-ready procurement and PLS seedbed MVP. The diagram shows the core actors and their business capabilities. The key redesign principle is that blockchain proof is a supporting use case, not the product's only use case.

X. UI/UX REVIEW AND REDESIGN PLAN

A. Current UX Diagnosis

The current UX problem is not only visual design. It is semantic design. The user must understand who they are, what organization they represent, what workflow state the case is in, what action is next, what is blocked, and whether proof is verified.

Observed or likely UX weaknesses:

- Dashboards risk becoming collections of cards rather than workflow entry points.
- Actor-specific routes are incomplete.
- Blockchain proof can appear detached from procurement context.
- Landing and dashboard experiences may not explain the full product story.
- Product copy can drift toward implementation language if not controlled.

B. Redesign Principles

- 1) **Case-centric design.** All actors work on the same canonical transaction case.
- 2) **Role-specific navigation.** Each actor sees only the workflows they own.
- 3) **Evidence panel pattern.** Audit and blockchain proof appear only on event, escrow, PLS distribution, or export contexts.
- 4) **Blocked-state clarity.** Pending review, suspended organization, forbidden action, and proof unavailable states must be explicit.
- 5) **Commercial copy discipline.** Use product words such as “Verify proof” and “Review escrow”, not “PBI coverage” or “workflow state resolver”.

C. Minimum Screen Set

XI. IMPLEMENTATION ROADMAP

A. Phase 0: Scope Lock and Evidence Reconciliation

- Reconcile backlog status against evidence.

Activity Diagram - Canonical Commercial Demo Case

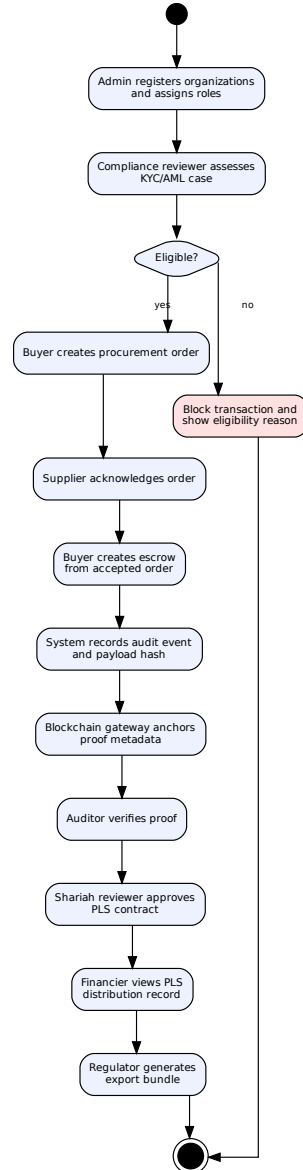


Fig. 2. Activity diagram for the recommended commercial demo case. A single case flows from governance and compliance into procurement, escrow, proof verification, PLS governance, and export.

- Ensure ‘backlog/backlog.csv’ is canonical.
- Update demo claims so they do not exceed implemented scope.

B. Phase 1: Actor Entry and Dashboards

- Add demo roles for administrator, supplier, compliance reviewer, Shariah reviewer, financier, regulator, and security operator.
- Extend dashboard resolver and navigation for all mandatory actors.
- Validate role access and unauthorized routes.

C. Phase 2: Administrator, Procurement, and Compliance

- Complete administrator member and role management UI.
- Complete buyer order and supplier acknowledgement workflow.
- Complete compliance case review and downstream eligibility gating.

D. Phase 3: Escrow and Proof Hardening

- Link escrow creation to accepted orders or an explicit demo order reference.

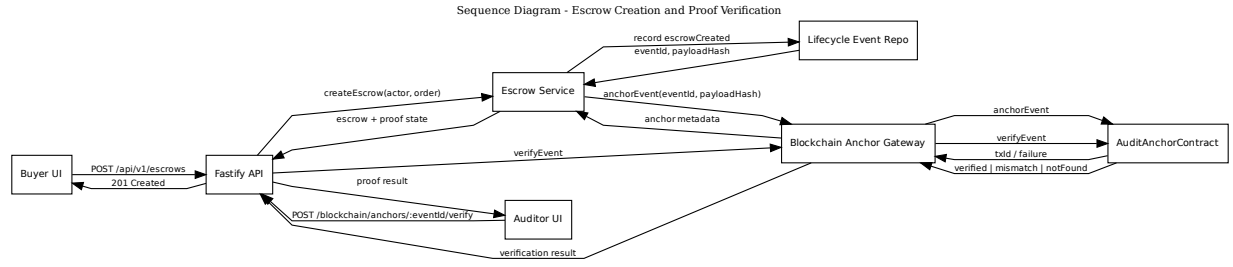


Fig. 3. Sequence diagram for escrow creation and blockchain proof verification. The sequence preserves the architectural rule that business events remain valid even if anchoring fails.

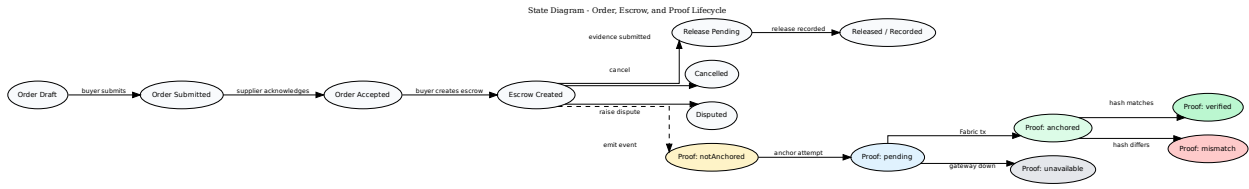


Fig. 4. State diagram for order, escrow, and proof lifecycle. Proof state is separated from business state so unavailable or mismatched proof never appears as released business value.

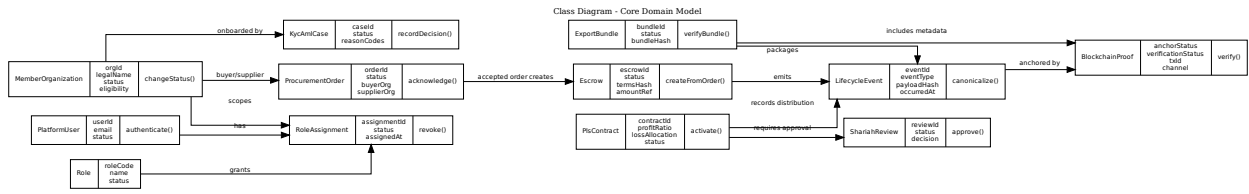


Fig. 5. Class diagram for the proposed domain model. The model centers on member organizations, roles, onboarding cases, procurement orders, escrow, lifecycle events, blockchain proofs, PLS contracts, and export bundles.

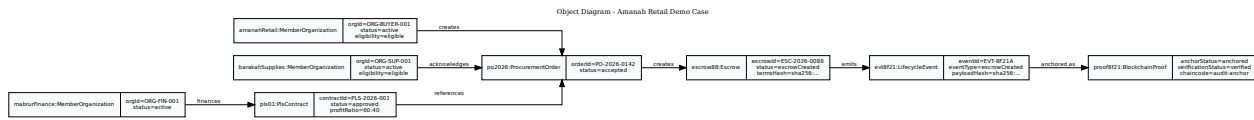


Fig. 6. Object diagram for the Amanah Retail demonstration case. Realistic sample data should be seeded so reviewers can understand how the product behaves across actors.

- Keep escrow proof panel honest across anchored, pending, failed, and unavailable states.
- Record lifecycle events for order acceptance and escrow creation.

E. Phase 4: Shariah and PLS

- Add Shariah review dashboard and PLS review detail.
- Implement PLS contract activation gate.
- Implement distribution record model and scenario tests.

F. Phase 5: Regulator Export and Release Evidence

- Add export request and bundle detail screens.
- Generate bundle manifest and integrity metadata.
- Create UAT scripts, final release evidence, and supervisor demo script.

XII. RISK ANALYSIS

XIII. SUCCESS METRICS

Recommended MVP success metrics are shown in Table IV.

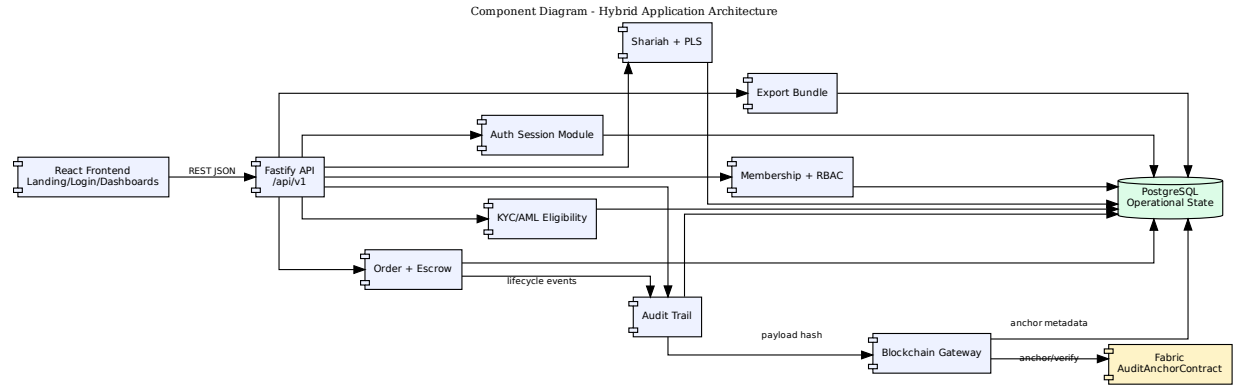


Fig. 7. Component diagram for the hybrid application architecture. PostgreSQL stores operational state, while Fabric stores selective proof metadata through a backend gateway.

TABLE II
MINIMUM SCREEN SET FOR ACTOR-COMPLETE MVP

Area	Required Screens
Public	Landing, Sign in
Shared	Dashboard, Settings, Logout, No Role, Pending Review, Forbidden, Service Unavailable
Administrator	Members, Member Detail, Roles, Role Assignment, Access History
Buyer	Orders, Order Detail, Escrow, Escrow Detail, Blockchain Proof
Supplier	Received Orders, Order Detail, Delivery Evidence, Escrow Status
Compliance	KYC Review, Case Detail, Eligibility Status
Shariah	Shariah Review, PLS Review Detail, Checklist, Decision
Financier	Financing, PLS Contract Detail, Distribution Detail
Auditor	Audit Trail, Event Detail, Blockchain Proof Verification
Regulator	Export Bundle, Export Detail, Verify Bundle
Operator/Developer	Runbook, API Quickstart, Environment Checklist

TABLE III
MAJOR PRODUCT, TECHNICAL, AND COMMERCIAL RISKS

Risk	Impact	Mitigation
Actor incompleteness	Product appears like a partial demo	Complete one workflow per mandatory actor before adding new infrastructure.
Blockchain overclaiming	Loss of credibility	Say “proof anchoring”, not “full production blockchain settlement”.
PLS overclaiming	Shariah/compliance risk	Require approval gate, deterministic scenarios, and limitation notes.
Data privacy leak	Compliance and trust failure	Keep raw KYC, payment, and commercial terms off-chain and out of broad UI.
Fabric demo fragility	Demo failure	Keep Fabric build/test as baseline and live network as optional unless fully scripted.
Runtime persistence gap	Demo data loss	Add PostgreSQL runtime composition switch.
UX semantic drift	Users cannot tell what to do	Use case-centric navigation and actor-specific pages.

XIV. FINAL RECOMMENDATION

The product solves a real market problem if it is narrowed to compliance-first procurement evidence and SME financing readiness. The current technical foundation is promising, but the product is not commercially ready until all mandatory actors can complete a coherent workflow. The next delivery should not prioritize more blockchain features. It should prioritize actor completeness, case-centric UX, eligibility enforcement, Shariah/PLS governance visibility, regulator export, runtime persistence, and release evidence.

The final MVP should be described as a *seedbed*: a credible, deployable, evidence-backed pilot that proves the architecture

and workflow. It should not be described as a production marketplace, full Islamic finance platform, or full consortium blockchain solution.

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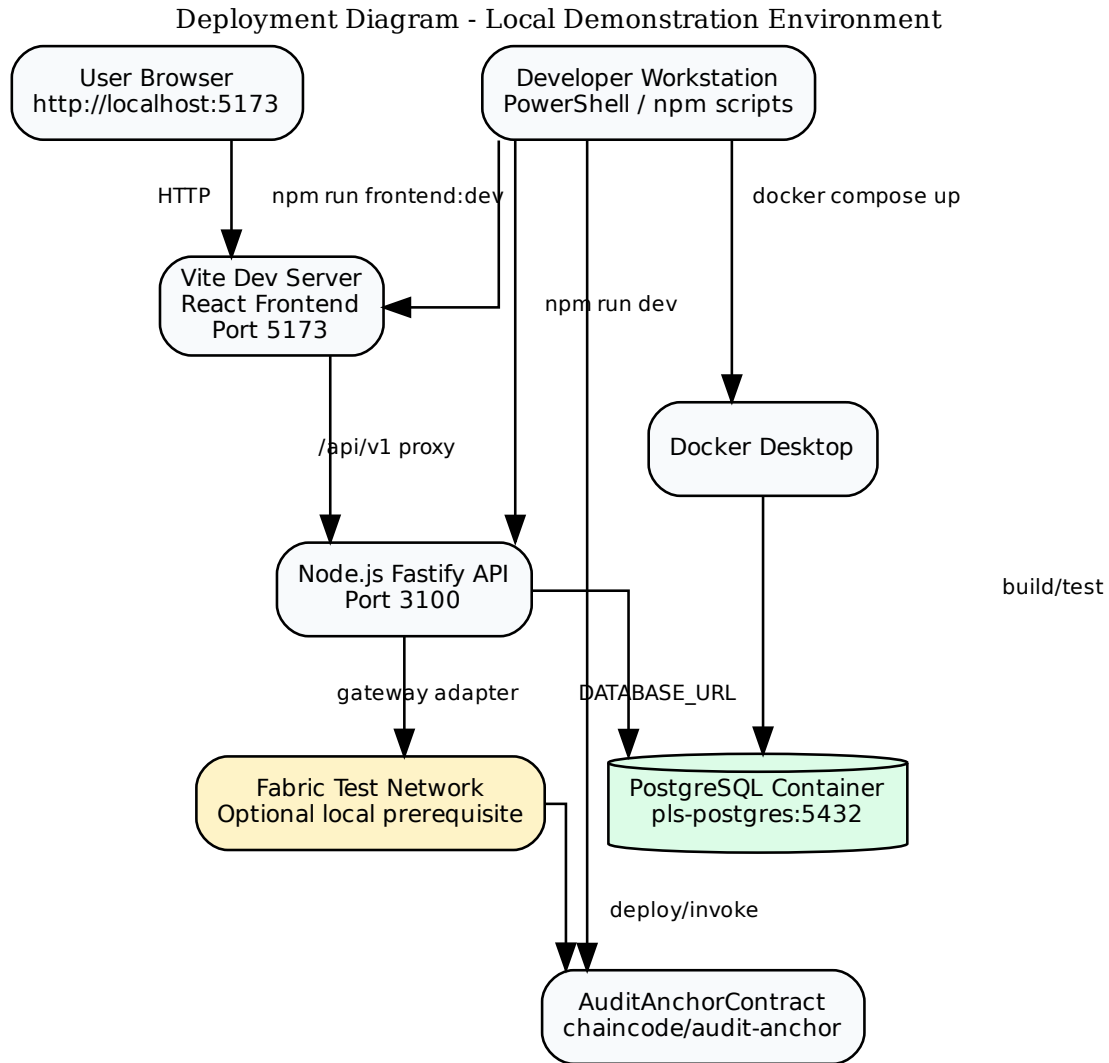


Fig. 8. Deployment diagram for the local demonstration environment. The target developer setup uses a Vite frontend, Fastify backend, PostgreSQL container, and optional Fabric test network.

TABLE IV
MVP SUCCESS METRICS

Metric	Target
Actor UAT completion	100% mandatory actors pass core path.
Authorization regression	0 critical unauthorized access defects.
Proof correctness	0 fake verified states; mismatch and unavailable clearly rendered.
Demo reproducibility	New operator can start local demo from runbook.
Evidence coverage	Every mandatory workflow has QA evidence.
Scope honesty	All post-MVP exclusions documented.
Time-to-demo	Supervisor case walkthrough under 20 minutes.

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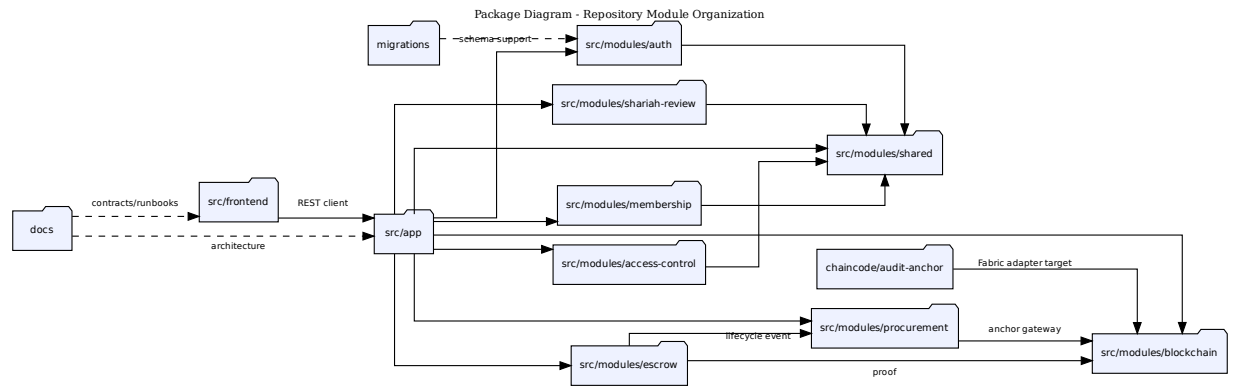


Fig. 9. Package diagram for repository module organization. The package structure should preserve clean boundaries between frontend, backend app, modules, migrations, chaincode, and documentation.

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